

**FACULTY OF ENGINEERING**  
**B.E. I - Semester (AICTE) (New) (Main) Examination, July 2021**

**Subject: Chemistry (Except Civil)**

Time: 2 Hours

Max. Marks: 70

- Note:** (i) First question is compulsory and answer any three questions from the remaining six questions.  
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
 (iii) Missing data, if any, may be suitably assumed.

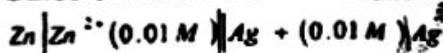
1 Answer any four questions.

(4 x 4 = 16 Marks)

- (a) Differentiate single electrode potential and standard electrode potential.  
 (b) Write the specifications of potable water.  
 (c) What is degree of polymerization?  
 (d) Name the monomers involved in the formation of Nylon 6:6 and Bakelite.  
 (e) Write the composition and uses of LPG.  
 (f) What do you mean by clean technology?  
 (g) Explain the relation between EMF and change in free energy.

(3x18 = 54 Marks)

2 (a) What is ion selective electrode? Explain construction and working of glass electrode.

(b) Calculate the  $E_{\text{cell}}$  of the following cell at 25°C by using Nernst equation.

$$(E^{\circ}_{\text{Ag}^{+}/\text{Ag}} = 0.801 \text{ V and } E^{\circ}_{\text{Zn}^{2+}/\text{Zn}} = -0.764 \text{ V})$$

3 (a) Define Alkalinity of water. Explain the estimation of alkalinity of water.

(b) Explain the mechanism of wet corrosion by (i) Evolution of hydrogen and  
 (ii) Absorption of oxygen.

4 (a) Discuss the applications of conducting polymers.

(b) Explain the preparation, properties and uses of silicone rubbers.

5 (a) Describe the composition and use of diesel fuel.

(b) Explain the fuel rating by octane number and cetane number.

6 (a) Describe the concept of trans-esterification and carbon neutrality for biodiesel.

(b) Explain the composition and characteristic properties of composites.

7 (a) Discuss construction and applications of methanol-oxygen fuel cell.

(b) What is biodegradable polymer? Explain properties and applications of polylactic acid.

\*\*\*\*

# FACULTY OF ENGINEERING

Code No. 15168/AICTE

B.E. I-Semester (AICTE) (New) (Main) Examination, July 2021

Subject: Engineering Physics

Time: 2 hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any three questions from the remaining six questions.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may suitably assumed.

1 Answer any four questions from the following:

(4x4=16 Marks)

- Define the term of 'Space lattice' and Unit cell.
- What are the applications of Hall effect?
- Define ionic polarization of a dielectric material.
- State the relationship between D, E, P
- Give few applications of Super conductors.
- What are the applications of Lasers?
- Define Numerical Aperture.

(3x18=54 Marks)

- Classify Crystal Imperfections.
  - Obtain an expression for concentration of Schottky defects in the case of ionic crystals.
- What are salient features of Kronig – Penney model?
  - Explain the formation of allowed and forbidden energy bands based on Kronig-Penney model.
- Deduce an expression for Electronic Polarizability.
  - Describe the experimental determination of dielectric constant of dielectric material by Schering Bridge method.
- Give the basic laws of Electricity and Magnetism.
  - Deduce Maxwell's Equations in differential form.
- Give an account of Weiss Molecular field theory of Ferro Magnetism.
  - Explain Type-I and Type-II super conductors
- Explain construction and working of Ruby Laser.
  - Discuss the Fibre drawing process (double crucible method).

.....

**FACULTY OF ENGINEERING**  
**B.E. I - Semester (AICTE) (New) (Main) Examination, July 2021**

**Subject: Environmental Science**

**Time: 2 Hours**

**Max. Marks: 70**

- Note:** (i) First question is compulsory and answer any three questions from the remaining six questions.  
(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
(iii) Missing data, if any, may be suitably assumed.

**Note: Answer any four questions.**

**(4x4 = 16 Marks)**

- 1 (a) List the issues that warrant the need for public awareness in environmental studies  
(b) Define ecological succession with an example  
(c) List the various techniques of ex-situ conservation of plant species.  
(d) List the various techniques for disposal of solid waste.  
(e) Define watershed management.  
(f) List the salient features of air act.  
(g) List the basic principles of disaster mitigation

**(3x18 = 54Marks)**

- 2 (a) Describe the importance of environmental studies.  
(b) Discuss the effects of modern agricultural system environment.
- 3 (a) Sketch the single channel flow of energy in an ecosystem.  
(b) Describe the various ecological pyramids.
- 4 (a) Discuss the various levels of biodiversity with the help of a neat sketch.  
(b) Discuss the various values of biodiversity with the help of a neat sketch.
- 5 (a) Define water pollution and discuss the various measures to control it.  
(b) Discuss the bag filter to control measures of air pollution with the help of a sketch.
- 6 (a) Discuss the causes, effects and control measures of Ozone layer depletion  
(b) Discuss the causes and effects of acid rain with the help of an example.
- 7 (a) Discuss the various types of disasters in detail.  
(b) Discuss about climate change & how it effects on human beings.

....

<https://www.osmaniaonline.com>

**Whatsapp @ 9300930012**

**Send your old paper & get 10/-**

**अपने पुराने पेपर्स भेजें और 10 रुपये पायें,**

**Paytm or Google Pay से**

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**

**B.E. (AICTE) I – Semester (Common for All Branches) (Main & Backlog)**  
**Examinations, March / April 2022**  
**Subject: Mathematics - I**

**Time: 3 Hours****Max. Marks: 70**

- Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each Question carries 14 Marks.  
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
 (iii) Missing data, if any, may be suitably assumed.

1.

- (a) Discuss the convergence of the series  $\sum \frac{1}{n^2}$ .  
 (b) Obtain the fourth degree Taylor's polynomial approximation to  $f(x) = e^{2x}$  about  $x = 0$ .  
 (c) Show that the following function  $f(x, y)$  is continuous at the point  $(0, 0)$ .  

$$f(x, y) = \begin{cases} \frac{2x(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$
  
 (d) Evaluate the double integral  $\iint_R e^{x^2} dx dy$ , where the region  $R$  is given by  $R: 2y \leq x \leq 2$  and  $0 \leq y \leq 1$ .  
 (e) Find the directional derivative of  $f(x, y, z) = xy^2 + 4xyz + z^2$  at the point  $(1, 2, 3)$  in the direction of  $3i + 4j - 5k$ .  
 (f) If  $\mathbf{r} = xi + yj + zk$  and  $r = |\mathbf{r}|$  show that  $\text{div} \left( \frac{\mathbf{r}}{r^3} \right) = 0$ .

2. (a) Examine the convergence or divergence of the following series:  $\sum \frac{x^{n+1}}{(n+1)\sqrt{n}}$

- (b) Test the convergence of the series  $\sum \frac{(-1)^{n-1}}{(2n-2)!}$ .

3. (a) Obtain the Taylor's polynomial approximation of degree  $n$  to the function

$$f(x) = e^x \quad \text{about the point } x = 0.$$

- (b) Using Lagrange mean value theorem, show that  $1 + x < e^x < 1 + xe^x$ .

4. (a) If  $f(x, y) = \tan^{-1}(xy)$ , find an approximate value of  $f(1.1, 0.8)$  using the Taylor's series (i) linear approximation and (ii) quadratic approximation.

- (b) Find the shortest distance between the line  $y = 10 - 2x$  and the ellipse

$$\frac{x^2}{4} + \frac{y^2}{9} = 1.$$

5. (a) Evaluate the integral  $\iiint_R (x^2 y) dx dy dz$ , where the boundary

$$R: x^2 + y^2 \leq 1, 0 \leq z \leq 1.$$

- (b) Evaluate the integral  $\iiint_R (2x - y - z) dx dy dz$ , where the boundary

$$R: 0 \leq x \leq 1, 0 \leq y \leq x^2, 0 \leq z \leq x + y.$$

-2-

6. (a) Find the work done by the force  $F = (x^2 - y^3)i + (x + y)j$  in moving a particle along the closed path  $C$  containing the curves  $x + y = 0$ ,  $x^2 + y^2 = 16$  and  $y = x$  in the first and fourth quadrants.
- (b) Let  $D$  be the region bounded by the closed cylinder  $x^2 + y^2 = 16$ ,  $z = 0$  and  $z = 4$ . Verify the divergence theorem if  $v = 3x^2i + 6y^2j + zk$ .
7. (a) Evaluate the surface integral  $\iint_S F \cdot n \, dA$  where  $F = 6z i + 6j + 3yk$  and  $S$  is the portion of the plane  $2x + 3y + 4z = 12$ , which is in the first octant.
- (b) Evaluate the integral  $\iint_S (\nabla \times v) \cdot n \, dA$  by Stoke's theorem where  $v = (x^2 - y^2)i + (y^2 - x^2)j + zk$  and  $S$  is the portion of the surface  $x^2 + y^2 - 2by + bz = 0$ ,  $b$  constant, whose boundary lies in the  $x$ - $y$  plane.

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**  
**B.E. I - Semester (AICTE) (Main) (New) Examination, July 2021**

**Subject: Programming for Problem Solving**

Time: 2 Hours

Max. Marks: 70

- Note:** (i) First question is compulsory and answer any three questions from the remaining six questions.  
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
 (iii) Missing data, if any, may be suitably assumed.

1 Answer any four questions.

(4 × 4 = 16 Marks)

- (a) Differentiate compiler and interpreter.
- (b) What is self-referential structure and given example?
- (c) What is a file? What are file operations?
- (d) In what way does an array differ from an ordinary variable?
- (e) Write the algorithm for linear search.
- (f) Define string. List any four string manipulation functions.
- (g) Write a function to find the sum of digits of a given number.

(3 × 18 = 54 Marks)

- 2 (a) Draw a flowchart to find the root of a quadratic equation.  
 (b) Explain about computer components in detail.
- 3 (a) Explain different ways of passing arguments to function with example.  
 (b) Write a C program to add the prime numbers of a certain range (0 to 10).
- 4 (a) Explain how arrays are passed to a function with an example.  
 (b) Write a program to find the second maximum in an array using function.
- 5 (a) How is a structure data type different from an array? Explain with an example.  
 (b) Write a program to display the prime numbers in a Fibonacci series using recursion.
- 6 (a) Why pointers should have data types when their size is always 4 bytes (in a 32-bit machine), irrespective of the variable they are pointing to?  
 (b) Write a program to copy contents from one existing file into another file.
- 7 (a) Write a short notes on call by reference.  
 (b) Explain linear search algorithm with suitable example.

**OSMANIA UNIVERSITY**  
**FACULTY OF ENGINEERING**  
**UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)**  
**B.E. ( All Branches ) II-Semester (Main) Examinations, October 2021**

**ENGINEERING MATHEMATICS-II**

**Time : 2 hours**

**Max. Marks : 70**

- Note :** i) *First Question is compulsory which carries 16 marks and it consists of 7 sub questions (a-g) and answer any 4 sub questions.*  
 ii) *Answer any Three questions from the remaining six questions (2-7). Each question carries 18 Marks.*  
 iii) *Missing data, if any, may suitably be assumed.*

|                                                                                                                                                                                | Marks | BT | CO |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 1. a) Find the rank of the matrix $A = \begin{bmatrix} 0 & 1 & -3 \\ 1 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix}$                                                                     | 4     | 1  | 1  |
| b) Check whether the following vectors are linearly dependent or not. If so, find the relation between them.<br>$x_1 = (1, 3, 4, 2), x_2 = (3, -5, 2, 2), x_3 = (2, -1, 3, 2)$ | 4     | 4  | 1  |
| c) Find the orthogonal trajectories of the family of curves $y = ce^x$ , $c$ is parameter.                                                                                     | 4     | 2  | 2  |
| d) Show that $x, x^2, x^3$ are linearly independent on any interval $I$ . <a href="https://www.osmaniaonline.com">https://www.osmaniaonline.com</a>                            | 4     | 1  | 3  |
| e) Find the constants $a, b$ such that the function $f(z) = x - 2ay + i(bx - ay)$ is analytic.                                                                                 | 4     | 5  | 4  |
| f) Expand $f(z) = \frac{1}{z}$ about $z = 2$ in Taylor's series.                                                                                                               | 4     | 6  | 5  |
| g) Solve $y'' + 2y' + 2y = 0$                                                                                                                                                  | 4     | 3  | 3  |
| 2. a) Solve the system of equations<br>$4x + 2y + z + 3w = 0$<br>$6x + 3y + 4z + 7w = 0$<br>$2x + y + w = 0$                                                                   | 9     | 6  | 1  |
| b) Using Cayley-Hamilton theorem, find $A^8$ , if $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$                                                                          | 9     | 3  | 1  |
| 3. a) Solve $y' + 4xy + xy^3 = 0$ .                                                                                                                                            | 9     | 3  | 2  |
| b) Find the general solution of the differential equation $y' = y^2 - (2x - 1)y + x^2 - x + 1$ , if $y = x$ is solution of the equation.                                       | 9     | 5  | 2  |

- |       |                                                                                                                                                 |   |   |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 4. a) | Using the method of variation of parameters.<br>solve $y'' + y = \operatorname{cosec} x$                                                        | 9 | 4 | 3 |
| b)    | Solve $(D^2 - 4)y = 2\cos^2 x$ .                                                                                                                | 9 | 6 | 3 |
| 5. a) | Evaluate $\int \frac{z^2+1}{z(2z-1)} dz$ , $C:  z  = 1$ , using Cauchy's integral formula.                                                      | 9 | 5 | 4 |
| b)    | Show that $u(x, y) = 2x + y^3 - 3x^2y$ is harmonic, find its harmonic conjugate and corresponding analytic function $f(z)$ .                    | 9 | 2 | 4 |
| 6. a) | Expand $f(z) = \frac{1}{z^2-3z+2} dz$<br>in the region (i) $0 <  z-1  < 2$ (ii) $1 <  z  < 2$                                                   | 9 | 4 | 5 |
| b)    | Find the Bilinear Transformation which maps the points $-1, 0, 1$ in the $z$ -plane onto the points $1, i, -1$ in the $w$ -plane.               | 9 | 6 | 5 |
| 7. a) | Evaluate the integral $I = \int_0^{2\pi} \frac{d\theta}{1-2a\cos\theta+a^2}$ where $a$ is the complex constant and (i) $ a  < 1$ (ii) $ a  > 1$ | 9 | 5 | 4 |
| b)    | Solve $2x^2y'' + 3xy' - 3y = x^3$                                                                                                               | 9 | 3 | 3 |

\*\*\*\*\*

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

**OSMANIA UNIVERSITY**  
**FACULTY OF ENGINEERING**  
**UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)**  
**B.E. (All Branches) II-Semester (Main) Examinations**  
**August/September 2022**

**ENGINEERING MATHEMATICS-II**

Time : 3 hours

Max. Marks : 70

- Note :** i) Answer Question No. 1 (Compulsory) and answer any four questions from the remaining questions (2- 7).  
 ii) Answers must be written in same order as they occur in the Question Paper.  
 iii) Missing data, if any, may suitably be assumed.

|                                                                                                                                                                                             | Marks | BT | CO |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 1. a) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$                                                                                     | 2     | 4  | 1  |
| b) Solve $(hx + by + f)dy + (ax + hy + g)dx = 0$                                                                                                                                            | 2     | 3  | 2  |
| c) Find the Particular Integral of $(D^2 + 6D + 9)y = 5e^{3x}$                                                                                                                              | 2     | 2  | 3  |
| d) Determine whether $\frac{1}{z}$ is analytic or not.                                                                                                                                      | 2     | 5  | 4  |
| e) For the conformal transformation $w = z^2$ , find the coefficient of magnification at $z = 2 + i$                                                                                        | 2     | 1  | 5  |
| f) Define Cauchy's Residue Theorem.                                                                                                                                                         | 2     | 6  | 5  |
| g) Find Integrating factor of $(2x \log x - xy)dy + 2y dx = 0$                                                                                                                              | 2     | 5  | 2  |
| 2. a) Find the values of k such that the system of equations $x + ky + 3z = 0$ , $4x + 3y + kz = 0$ , $2x + y + 2z = 0$ has non-trivial solution.                                           | 7     | 4  | 1  |
| b) Reduce the following Quadratic form into "Sum of squares" by an orthogonal transformation and give the matrix of transformation $x^2 + 2y^2 - 7z^2 - 4xy + 8xz$ and discuss with nature. | 7     | 3  | 1  |
| 3. a) Find the orthogonal trajectory of family of curves $r^n \sin n\theta = a^n$                                                                                                           | 7     | 4  | 2  |
| b) Solve $y' + y \tan x = \cos x$ , $y(0) = 0$                                                                                                                                              | 7     | 5  | 2  |

(P.T.O.)

4. a) Solve  $(x^2 D^2 - xD - 3)y = x^2 \log x$  7 2 3
- b) Using the Method of Variation of Parameters solve  $\frac{d^2 y}{dx^2} + 4y = \tan 2x$  7 3 3
5. a) Use Cauchy's Integral formula to evaluate  $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ , where C is the circle  $|z| = 3$  7 5 4
- b) Show that  $e^x(x \cos y - y \sin y)$  is a harmonic function. Find the analytic function for which  $e^x(x \cos y - y \sin y)$  is imaginary part. 7 1 4
6. a) Find the bilinear transformation which maps the points  $z = 1, i, -1$  into the points  $w = i, 0, -i$  7 6 5
- b) Find the residue of  $\frac{z^3}{(z-1)^4(z-2)(z-3)}$  at a pole of order 4 7 4 5
7. a) Evaluate  $\int_C \frac{e^{2z}}{(z+1)^4} dz$  around  $c: |z-1| = 3$  7 3 4
- b)  $\int_C \frac{1-2z}{z(z-1)(z-2)} dz$ , where c is the circle  $|z|=1.5$  by using Cauchy's Residue Theorem. 7 6 5

\*\*\*\*\*

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

**OSMANIA UNIVERSITY**  
**FACULTY OF ENGINEERING**  
**UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)**  
**B.E. (All Branches) II-Semester (Main) Examinations**  
**August/September 2022**

**ENGINEERING MATHEMATICS-II**

Time : 3 hours

Max. Marks : 70

- Note :** i) Answer Question No. 1 (Compulsory) and answer any four questions from the remaining questions (2- 7).  
 ii) Answers must be written in same order as they occur in the Question Paper.  
 iii) Missing data, if any, may suitably be assumed.

|                                                                                                                                                                                             | Marks | BT | CO |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|
| 1. a) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$                                                                                     | 2     | 4  | 1  |
| b) Solve $(hx + by + f)dy + (ax + hy + g)dx = 0$                                                                                                                                            | 2     | 3  | 2  |
| c) Find the Particular Integral of $(D^2 + 6D + 9)y = 5e^{3x}$                                                                                                                              | 2     | 2  | 3  |
| d) Determine whether $\frac{1}{z}$ is analytic or not.                                                                                                                                      | 2     | 5  | 4  |
| e) For the conformal transformation $w = z^2$ , find the coefficient of magnification at $z = 2 + i$                                                                                        | 2     | 1  | 5  |
| f) Define Cauchy's Residue Theorem.                                                                                                                                                         | 2     | 6  | 5  |
| g) Find Integrating factor of $(2x \log x - xy)dy + 2y dx = 0$                                                                                                                              | 2     | 5  | 2  |
| 2. a) Find the values of k such that the system of equations $x + ky + 3z = 0$ , $4x + 3y + kz = 0$ , $2x + y + 2z = 0$ has non-trivial solution.                                           | 7     | 4  | 1  |
| b) Reduce the following Quadratic form into "Sum of squares" by an orthogonal transformation and give the matrix of transformation $x^2 + 2y^2 - 7z^2 - 4xy + 8xz$ and discuss with nature. | 7     | 3  | 1  |
| 3. a) Find the orthogonal trajectory of family of curves $r^n \sin n\theta = a^n$                                                                                                           | 7     | 4  | 2  |
| b) Solve $y' + y \tan x = \cos x$ , $y(0) = 0$                                                                                                                                              | 7     | 5  | 2  |

(P.T.O.)

4. a) Solve  $(x^2 D^2 - xD - 3)y = x^2 \log x$  7 2 3
- b) Using the Method of Variation of Parameters solve  $\frac{d^2 y}{dx^2} + 4y = \tan 2x$  7 3 3
5. a) Use Cauchy's Integral formula to evaluate  $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ , where C is the circle  $|z| = 3$  7 5 4
- b) Show that  $e^x(x \cos y - y \sin y)$  is a harmonic function. Find the analytic function for which  $e^x(x \cos y - y \sin y)$  is imaginary part. 7 1 4
6. a) Find the bilinear transformation which maps the points  $z = 1, i, -1$  into the points  $w = i, 0, -i$  7 6 5
- b) Find the residue of  $\frac{z^3}{(z-1)^4(z-2)(z-3)}$  at a pole of order 4 7 4 5
7. a) Evaluate  $\int_C \frac{e^{2z}}{(z+1)^4} dz$  around  $c: |z-1| = 3$  7 3 4
- b)  $\int_C \frac{1-2z}{z(z-1)(z-2)} dz$ , where c is the circle  $|z|=1.5$  by using Cauchy's Residue Theorem. 7 6 5

\*\*\*\*\*

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

**FACULTY OF ENGINEERING****B.E. II – Semester (AICTE) (Main & Backlog) Examinations, September/October-2022****Subject: English****Max. Marks: 70****Time: 3 Hours**

**Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each Question carries 14 Marks.  
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
 (iii) Missing data, if any, may be suitably assumed.

**1. a) Correct the following sentences.**

i. Mr. Smith is a European.

ii. There is a temple besides my house.

**b) Choose the correct alternative to fill in the blank.**

i. You are not eligible for a subsidized bus pass \_\_\_\_\_ you are a full-time student. (unless/ moreover/ although)

- | ii. The train was delayed \_\_\_\_\_ bad weather. (in view of / because of / resulting)

**c) Convert the passage into Passive voice.**

My uncle painted this picture. Someone has offered him a lot of money for it. He will deliver the painting tomorrow.

**d) Convert the sentences into reported speech.**

i. I said to her, "Please come early."

ii. The principal said, "What were you doing when I saw you?"

**e) Write the correct form of the verb supplied.**

i. Robin (sell) his car by next Sunday.

ii. At this time tomorrow morning, they (begin) working

**f) Give gender- neutral expressions for the following nouns.**

i. Housewife

ii. Freshman

iii. Negro

iv. Policeman

**g) Give the equivalents of the following.**

i. emoticon \_\_\_\_\_

ii. education+ entertainment

iii. differently abled

iv. economically marginalized

2/ a) Read the passage and answer the questions that follow.

His wife lit the domestic fire at dawn, boiled water in a mud pot, threw into it a handful of millet flour, added salt, and gave him his first nourishment for the day. When he started out, she would put in his hand a packed lunch, once again the same millet cooked into a little ball, which he could swallow with a raw onion at midday. She was old, but he was older and needed all the attention she could give him in order to be kept alive.

(i) What did Muni's wife do early in the morning?

(ii) After boiling the water in a mud pot, she added some rice.

(TRUE / FALSE)

(iii) Give the one-word substitute of the phrase 'the food necessary for growth, health, and good condition'.

(iv) Why does she give so much attention to him?

b) Write a paragraph in about 200 words from the given hint and give a suitable title

Junk food --- nutritional value --- is marketed, --- healthy. --- are high in saturated fats --- sugar components --- excess of salts --- lack any fiber. --- popularity --- ready to eat --- cook.

3. a) "If-" contains a multitude of characteristics deemed essential to the ideal man. List them in 150 words

b) Explain the following processes of word formation in English, with adequate examples

i. Compounding

ii. Contractions

4. a) Read the passage and answer the questions that follow.

I say to you today, my friends, so even though we face the difficulties of today and tomorrow I still have a dream. It is a dream deeply rooted in the American dream. I have a dream that one day this nation will rise up and live out the true meaning of its creed: "We hold these truths to be self-evident, that all men are created equal."

(i) What do you mean by the "American Dream"?

(ii) What is the meaning of 'creed'?

(iii) What difficulties were faced by the Negroes?

(iv) What is Martin Luther King's dream?

b) Write a letter requesting for Internship to the HR department, Leverage solutions, Ltd, Hyderabad, stating why you should be hired for the internship and how you will be benefiting the organisation.

5. a) Read the lines and answer the questions that follow.

I shall be telling this with a sigh  
Somewhere ages and ages hence:  
Two roads diverged in a wood, and I—  
I took the one less traveled by,  
And that has made all the difference.

- (i) Give the synonym of 'sigh'.  
(ii) What does the poet refer to when he says 'ages and ages hence'?  
(iii) Why does the poet repeat the lines 'Two roads.....less traveled by,' in the fourth stanza?  
(iv) What is the message given by the poet?

- b) What is a formal report? Describe its structure and components.

6. a) Read the passage and answer the questions that follow.

Dr. Arnold, generally regarded as the founder of the modern public school, looked on games as simply a waste of time. Then, chiefly in England and the United States, games were built up into a heavily-financed activity, capable of attracting vast crowds and rousing savage passions, and the infection spread from country to country. It is the most violently combative sports, football and boxing, that have spread the widest.

- (i) What is Dr. Arnold's opinion about games?  
(ii) Give the synonym of 'vast'.  
(iii) According to the passage, games were built up into a heavily-financed activity in Russia. (True / False).  
(iv) Which two sports are mentioned by the author as the most violently combative sports?

- b) What is an SoP? Briefly describe its objective, form and structure.

- ✓ 7. a) Read the passage and answer the questions that follow.

Marie Curie was one of the most accomplished scientists in history. Together with her husband, Pierre, she discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substances. Pierre and Marie's amicable collaboration later helped to unlock the secrets of the atom.

Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics.

-4-

Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of whom was Pierre Curie. Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marie was stunned by this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress.

Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first woman to be given a professorship at the world-famous university. In 1911, she received the Nobel Prize in chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusioned about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world.

I. The Curies' \_\_\_\_\_ collaboration helped to unlock the secrets of the atom.

i.friendly ii.competitive iii.courteous iv.industrious

II. Marie had a bright mind and a \_\_\_\_\_ personality.

i.strong ii.lighthearted iii.humorous iv.strange

III. When she learned that she could not attend the university in Warsaw, she felt \_\_\_\_\_.

i.hopeless ii.annoyed iii.depressed iv.worried

IV. Marie \_\_\_\_\_ by leaving Poland and traveling to France to enter the Sorbonne.

i.challenged authority ii.showed intelligence iii.behaved  
iv.was distressed

V. \_\_\_\_\_ she remembered their joy together.

i.Dejectedly ii.Worried iii.Tearfully iv.Happily

VI. Her \_\_\_\_\_ began to fade when she returned to the Sorbonne to succeed her husband.

i.misfortune ii.anger iii.wretchedness iv.disappointment

..5..

VII. Even though she became fatally ill from working with radium,

1. Marie Curie was never \_\_\_\_\_.

i. troubled ii. worried iii. disappointed iv. sorrowful

✓b) Rearrange the following sentences to make a meaningful paragraph.

- A. While their trappings and forms and authority are sought to be zealously protected, their spirit and substance have been steadily slipping.
- B. Perceived to be corrupt, arrogant, self serving and insensitive, the executive has become the butt of ridicule
- C. Our parliamentary democracy has come under heavy stress and strain often leading us to wonder what is becoming of our hallowed institutions of governance.
- D. The Legislature, the Executive and the Judiciary have been part of the process of decay often competing with each other in destroying the spirit of parliamentary democracy.
- E. Perpetuating unconscionable delays in administering justice caught in archaic processes of its own making and turf war.
- F. The judiciary too has lost much of its majesty.
- G. The presence of alleged bandits and criminals besmirch our legislative bodies and erode their credibility.

i. EFCADBG ii. CDEGABG iii. ADBCEFG iv. CADFBEG

..

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**  
**B.E. II - Semester (AICTE) (Main) Examination, October 2021**

**Subject: Mathematics - II**

Time: 2 Hours

Max. Marks: 70

- Note: i) First Question is compulsory and answer any three questions from the remaining six questions.  
 ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.  
 iii) Missing data, if any, may suitably be assumed.

Answer any four questions from the following.

**(4x4=16 Marks)**

- 1 a Find the rank of the matrix  $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & 3 & -1 \\ 8 & 13 & 14 \end{bmatrix}$ .
- b Solve  $y(2xy + e^x)dx = e^x dy$ .
- c Solve  $(D^2 + 9)y = \sin 3x$ .
- d Evaluate  $\int e^{-4x}(1 - e^{-x})^2 dx$  in terms of beta function.
- e Find  $L\{t^3 e^t + \sin^2 t\}$ .
- f Find  $L^{-1}\left\{\frac{1}{(x^2+1)(x^2+3)}\right\}$ .
- g Evaluate  $6P_3(x) + 4P_1(x) - 16P_1(x)$  as a polynomial of  $x$ .

**(3x18=54 Marks)**

- 2 (a) Find the eigen values and eigen vectors of the matrix  $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$ .

(b) Reduce the quadratic form  $8x_1^2 + 7x_2^2 + 3x_3^2 + 12x_1x_2 + 4x_1x_3 - 8x_2x_3$  into canonical form.

- 3 (a) Solve  $y(x+y)dx - x^2 dy = 0$ .  
 (b) Solve  $y(2xy+1)dx + x(1+2xy-x^2y^2)dy = 0$ .

- 4 (a) Solve  $y' + 4y = x \cos x$ .  
 (b) Solve  $y' + 2y + y = e^{-x} \log x$  by the method of variation of parameters.

- 5 (a) Find the power series solution of the differential equation  $y'' + 2xy' + y = 0$  about the origin.

..2..

(b) Evaluate  $\frac{d}{dx} [\operatorname{erf}(ax)]$ .

6 (a) Find  $L\left\{\int_0^t u e^{-u} \sin 4u \, du\right\}$ .

(b) Find  $L^{-1}\left\{\frac{1}{s^2(s+2)}\right\}$ .

7 (a) Find the orthogonal trajectories of the family of curves  $y' + 3x^c = c$  where  $c$  is arbitrary constant.

(b) Solve  $x^2 y' - xy' - 3y = x^2 \log x$ .

....

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजें और 10 रुपये पायें,

Paytm or Google Pay से

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**  
**B.E. II - Semester (AICTE) (Main & Backlog) New) Examination,**  
**September/ October - 2022**

**Subject : MATHEMATICS-II**

**Time : 3 Hours**

**Max. Marks: 70**

- Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each Questions carries 14 Marks.  
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.  
 (iii) Missing data, if any, may be suitably assumed.

1. (a) If  $\lambda$  is an eigenvalue of a non-singular matrix  $A$ , show that  $\frac{|A|}{\lambda}$  is an eigenvalue of

$\text{Adj } A$ .

(b) Obtain the general solution of the differential equation  $y = xy' + e^{-y'}$ .

(c) Find the second order differential equation for which  $e^x, e^{-x}$  are solutions.

(d) Prove that  $\text{erf}(x) + \text{erfc}(x) = 1$ .

(e) Find  $L\{(\cos t - \sin t)^2\}$ .

(f) Find the matrix of the quadratic form  $Q = 2(x^2 + xy + y^2)$ .

(g) Find a particular integral of  $y'' + 2y' + y = \sin x$ .

2. (a) Show that the system of equations  $x - 3y - 8z + 10 = 0$ ,  $3x + y - 4z = 0$ ,  $2x + 5y + 6z - 13 = 0$  is consistent and solve the same

(b) Verify Cayley-Hamilton theorem for  $A = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ .

3. (a) Find the general solution of  $(x^3 + y^3) dx - xy^2 dy = 0$ .

(b) Solve the differential equation  $xy(1 + xy^2) \frac{dy}{dx} = 1$ .

-2-

4. ✓ (a) Solve  $\frac{d^3 y}{dx^3} - y = (e^x + e^{-x})^2$ .

✓ (b) Solve  $x^2 y'' - 2xy' + 2y = \frac{1}{x}$ . CE

5. (a) Prove that  $\beta(m, n) = \beta(n, m)$  and  $\beta(m+1, n) + \beta(n+1, m) = \beta(m, n)$ .

(b) Find the power series solution of the differential equation  $(1-x^2)y'' - 2xy' + 2y = 0$  about the origin.

6. (a) Evaluate  $\int_0^{\infty} t^3 e^{-t} \sin t \, dt$  using Laplace transform.

✓ (b) Apply convolution theorem to find  $L^{-1} \left\{ \frac{1}{s(s^2-1)} \right\}$ .

7. (a) Define rank of a matrix. Find all values of  $k$  such that the rank of the matrix

$A = \begin{pmatrix} k & -1 & 0 & 0 \\ 0 & k & -1 & 0 \\ 0 & 0 & k & -1 \\ -6 & 11 & -6 & 1 \end{pmatrix}$  is equal to 3.

✓ (b) Find the orthogonal trajectories of the family of curves  $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$ , where  $\lambda$  is a parameter.

\*\*

# FACULTY OF ENGINEERING

## B.E. (Except I.T.) III-Semester (CBCS) (Backlog) Examination, July 2021

Subject : Engineering Mathematics - III

Max. Marks: 70

Time: 2 hours

Note: Missing data, if any, may be suitably assumed.

### PART - A

(5x2 = 10 Marks)

Answer any five questions.

- 1 Determine 'p' such that  $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{px}{y}\right)$  is an analytic function.
- 2 Evaluate  $\oint_C \frac{e^{2z}}{(z+2)^4} dz$  where C is the circle  $|z| = 3$ .
- 3 Find the residue of  $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$  at its pole  $z = 2$ .
- 4 Classify singularity of the function  $f(z) = \frac{e^{1/z}}{z^2}$ .
- 5 State Dirichlet's conditions for the existence of Fourier series.
- 6 Find  $b_n$  in the Fourier series expansion of  $f(x) = 2x - x^2$  in  $(0, 3)$ .
- 7 Find the partial differential equation by eliminating the arbitrary function  $f$  from  $z = f(x^2 - y^2)$ .
- 8 Solve  $xp + yq = 3z$ .
- 9 Solve  $z = px + qy + \sqrt{1 + p^2 + q^2}$ .
- 10 Find a particular integral of the equation  $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^2 z}{\partial x^2 \partial y} + 4 \frac{\partial z}{\partial y} = e^{x+2y}$ .

### PART - B

(4x15 = 60 Marks)

Answer any four questions.

- 11(a) Find the analytic function  $f(z) = u + iv$  where  $v(x, y) = e^{-x}(x \cos y + y \sin y)$ .

- (b) If  $F(a) = \oint_C \frac{4z^2 + z + 5}{z - a} dz$  where C is the ellipse  $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ , then find,

(i)  $F(3.5)$  (ii)  $F(i)$  (iii)  $F'(-1)$  (iv)  $F''(-i)$

- 2 Find the Laurent's series expansion of  $f(z) = \frac{z^2 - 6z - 1}{(z-1)(z-3)(z+2)}$  in the region

$3 < |z+2| < 5$ .

(iii) Evaluate  $\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1}$ .

..2..

13 Obtain half range Fourier cosine series of the function

$$f(x) = \begin{cases} kx, & 0 \leq x \leq l/2 \\ k(l-x), & l/2 \leq x \leq l \end{cases}$$

Hence find the sum  $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

14 (a) Solve  $x(y-z)p + y(z-x)q = z(x-y)$ .

(b) Solve  $2xz - px^2 - 2qxy + pq = 0$  by Charpit's method.

15 Solve  $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$  with boundary conditions  $u(x, 0) = 3\sin\pi x$ , and  $u(0, t) = u(1, t) = 0$  where  $0 < x < 1, t > 0$ .

16 (a) Show that the bilinear transformation  $w = \frac{z+3}{z-4}$  maps the circle  $x^2 + y^2 - 4x = 0$  into the line  $4u + 3 = 0$ .

(b) Using Cauchy's residue theorem evaluate  $\oint_C \frac{z}{(z-1)(z-2)^2} dz$  where  $C$  is the circle  $|z-2| = \frac{1}{2}$

17 (a) Solve

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin x.$$

(b) Solve  $z^2 = pq$ .

\*\*\*\*\*

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पावें,

Paytm or Google Pay से

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**  
**BE III Semester (CBCS((Backlog) Examination, November 2021**

**Subject: Engineering Mathematics - III (Except I.T)**

**Time: 2 Hours**

**Max. Marks: 70**

(Missing data, if any, may be suitably assumed)

**PART - A**

**Note: Answer any five questions.**

**(5x2=10 Marks)**

1. Show that  $f(z) = xy + iy$  is everywhere continuous but is not analytic.
2. State Cauchy's integral formula for derivatives.
3. Show that  $f(z) = \frac{\sin z}{z}$  has removable singularity.
4. Expand  $f(z) = \frac{1}{z}$  about  $z = 2$  in Taylor's series.
5. Express  $f(x) = x$  as half-range sine series in  $0 < x < 2$ .
6. Write Fourier series expansion of even periodic function  $f(x)$  in  $(-c, c)$ .
7. Form the partial differential equation by eliminating the arbitrary function from  $z = (x + y)\phi(x^2 - y^2)$ .
8. Solve  $xp + yq = 3z$ .
9. Classify the partial differential equation  $\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ .
10. Solve  $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y}$  using the separation of variables.

**PART - B**

**Note: Answer any four questions.**

**(4x15=60 Marks)**

11. Find 'k' such that  $f(x, y) = x^3 - 3kxy^2$  is harmonic and find its harmonic conjugate.
12. Evaluate  $\oint_C \frac{e^{2z}}{(z+i)^3} dz$ ,  $C: |z| = 3$ , using Cauchy's integral formula.
13. a) Evaluate  $\oint_C \frac{ze^z}{(z^2+4)} dz$ , where  $C$  is  $|z| = 5$ .  
 b) Find the bilinear transformation which maps the points  $(1, i, -1)$  of  $z$ -plane to  $(2, i, -2)$  of  $w$ -plane.
14. Find the Fourier series expansion of following periodic function  $f(x)$  of period  $2\pi$   
 $f(x) = \begin{cases} 2+x, & -2 \leq x \leq 0 \\ 2-x, & 0 \leq x \leq 2 \end{cases}$  hence deduce that  $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$
14. a) Find the general solution of the partial differential equation  $(y+z)p + (z+x)q = x+y$ .  
 b) Solve  $(D^2 + 4DD' - 5D'^2)z = \sin(2x + 3y)$ .

..2..

15. An elastic string of length  $l$  which is fastened at its end  $x = 0$  and  $x = l$  is picked up at its center point  $x = l/2$  to a height of  $h$  and released from rest. Find the lateral displacement of the string at any instant time.

10. a) Expand  $f(z) = \frac{1}{(z^2 - 3z + 2)}$  as Laurent's series in the region

(i)  $0 < |z - 1| < 1$  (ii)  $0 < |z| < 2$ .

Solve  $2\sqrt{p} + 3\sqrt{q} = 6x + 2y$ .

17. a) Show that  $f(z) = \bar{z}$  is continuous at the point  $z = 0$  but not differentiable at  $z = 0$ .

Express  $f(x) = \frac{x}{2}$  as a Fourier series in  $-\pi < x < \pi$ .

\*\*\*\*

<https://www.osmaniaonline.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

<https://www.osmaniaonline.com>

**FACULTY OF ENGINEERING**

**B.E. (CE / EE / Inst. / CSE) (AICTE) IV – Semester (Main) Examination,**

**December 2020**

**Subject: Indian Constitution**

**Time: 2 Hours**

**Max.Marks: 70**

**PART – A**

**Note: Answer any five questions.**

**(5 x 2 = 10 Marks)**

- 1 Explain the main features of the Indian Councils Act 1909.
- 2 When and why the Constituent Assembly was formed?
- 3 Who appoints the Prime Minister of India? What are the functions of the Prime Minister?
- 4 What are Panchayati Raj Institutions? Name them.
- 5 List out the fundamental duties as enshrined in the Constitution of India.
- 6 What are Directive Principles? How are they useful?
- 7 Discuss the Centre-State Legislative Relations.
- 8 Explain the composition and functions of NITI Aayog.
- 9 Explain the functions of the Finance Commission of India.
- 10 Mention the composition and functions of the National Commission for Women.

**PART – B**

**Note: Answer any four questions.**

**(4 x 15 = 60 Marks)**

- 11 Examine the role of the Constituent Assembly in the making of the Indian Constitution.
- 12 Explain in detail the fundamental features of the Constitution of India.
- 13 Discuss the powers and functions of the President of India.
- 14 Critically analyze the role of Governor in the Administration of the State.
- 15 Explain in detail the fundamental rights as mentioned in the Constitution.
- 16 Examine the role of the Election Commission of India in conducting the elections.
- 17 Write short notes on:
  - a) Government of India Act, 1935.
  - b) Urban Government.
  - c) Inter-State Council.
  - d) National Human Rights Commission.

\*\*\*\*

**FACULTY OF ENGINEERING**  
**B.E. 2/4 (CSE) I Semester (Suppl.) Examination, July 2012**  
**BASIC ELECTRONICS**

Time: 3 Hours]

[Max. Marks : 75

**Note : Answer all questions from Part A. Answer any five questions from Part B.**

**PART – A****25**

1. What is Barkhausen criterion for oscillations ? 2
2. What are the characteristics of an ideal OPAMP ? 3
3. A diode acts as a switch. Justify. 2
4. Realize OR gate using NAND gates. 3
5. List out the applications of SCR. 2
6. Define Ripple factor of a rectifier. 2
7. Draw the symbols of n-p-n Transistor, n-channel FET and SCR. 3
8. What is an extrinsic semiconductor ? 3
9. Classify the amplifiers based on conduction angle. 3
10. What is a transducer ? 2

**PART – B****50**

11. Draw the circuit diagram of a full wave rectifier and explain its working principle. Derive expressions for the Ripple factor, Regulation and efficiency. 10
12. Explain the working of P-N junction diode with its V-I characteristics and mention the applications of P-N junction diode. 10



Code No. : 5344/S

13. a) Draw the input and output characteristics of CE transistor and discuss. 5  
b) Explain the working of zener voltage regulator with neat circuit diagram. 5
14. a) What are the disadvantages of negative feedback ? Explain how the input and output impedances of an amplifier are effected by the negative feedback. 5  
b) Draw the circuit of Colpitts or Hartley oscillator and explain its working. 5
15. How the operational amplifier acts as integrator and differentiator ? Explain. 10
16. What are the universal gates ? Design the half and full adder circuits using NAND gates. 10
17. Explain the working of CRO with neat block diagram. 10

## FACULTY OF ENGINEERING

B.E. 2/4 (CSE) I – Semester (Main) Examination, Nov./Dec. 2012

Subject: Computer Architecture

Time: 3 Hours

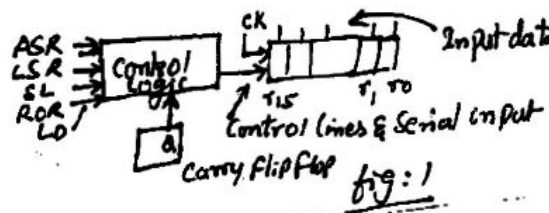
Max. Marks: 75

**Note: Answer all questions from Part A. Answer any five questions from Part B.****PART – A (25 Marks)**

- Write reverse polish notation for the following:  
 $(A+B) * [C * (D+E) + F]$  (2)
- Draw a block diagram for the 'add' micro-operation when implemented in a serial computer. (2)
- Explain arithmetic and divide overflow with some examples for 2'S complement numbers. (3)
- Why is bus arbitration required? Write any two bus-arbitration methods. (3)
- Write the IEEE 754 standard representation scheme for single and double precision floating point number. (3)
- What are the various strategies for parallel processing in a computer? (2)
- Explain Asynchronous communication interface. (2)
- What is the difference between isolated I/O and memory mapped I/O. Mention its advantages and disadvantages. (3)
- A 128 MB main memory has a 64 KB direct-mapped cache with 16 bytes per line. How many lines are there in Cache and show main memory address bits are partitioned. (3)
- What is Cache coherence? Mention the solutions to Cache coherence problem. (2)

**PART – B (50 Marks)**

- Illustrate various paths available in the control unit with a flow chart. Summarize the paths that controls tasks during various computer cycles. (10)
- An ALU of a processor uses shift register shown in the Fig. to perform shift and rotate operations. Inputs to the control logic for this register consist of: ASR: Arithmetic shift right, LSR: logic shift right, SL: shift left, RDR: rotate right, LD: parallel load. All shift and load operations are controlled by one clock input. The shift register is an edge triggered 'D' flip flop give the complete logic diagram for control logic for  $r_0$ ,  $r_1$  and  $r_{15}$ .



- Evaluate  $(A+B) * (C+D)$  using 3, 2, 1, 0 address instruction. (6)
  - What is the use of fast multipliers? Write about array multipliers. (4)
- Explain the sequence of events that takes place after an interrupt occurs through flow chart. (6)
  - Differentiate between cycle stealing and burst mode transfer of DMA. (4)

- 2 -

- 15.(a) Show how multiplication and division operations would be performed by hardware with a narrative flow chart. (6)
- (b) Explain vector loops in a vector processor. (4)
- 16.(a) A two level memory system has 8-virtual pages on disk to be mapped into four page frames in the main memory. A computer programme generates the following page trace: 2,2,1,0,7,6,1,0,5,6,6,3,2,3,7,6 (i) sketch the successive pages residing in four page frames with respect to above page tracing using LRU replacement policy. Compute hit ratio also. (6)
- (b) Show the hardware organization of associative memory and explain the read and write operation. (4)
17. Write short notes on:
- a) Floating Point Arithmetic (5)
- b) Array Processors. (5)

-:-:-

**FACULTY OF ENGINEERING**

**B.E. 4/4 (CSE) II-Semester (Main) Examination, April / May 2013**

**Subject : Advanced Databases  
(Elective-III)**

**Time : 3 Hours**

**Max. Marks: 75**

**Note: Answer all questions of Part - A and answer any five questions from Part-B.**

**PART – A (25 Marks)**

1. Define the Syntax for attributes and operations for classes in UML. (3)
2. Explain the Nested Queries. (2)
3. Explain built in constructed types. (3)
4. What is parallelism? (2)
5. What is the difference between XML DTD and XML schema? (3)
6. What is Data fragmentation? (2)
7. Define call-level-interface. (3)
8. Explain Recursive Associations. (3)
9. Define the ODMG ODL. (2)
10. Define Homogeneous and Heterogeneous Databases. (2)

**PART – B (5x10=50 Marks)**

- 11.(a) Explain about Persistent Stored modules.  
(b) With an example explain different components of E-R diagram.
- 12.(a) Explain inheritance of structured types in SQL with suitable examples.  
(b) Explain table hierarchies in detail.
- 13.(a) Explain major components of ODMG standard.  
(b) Explain about XML schema with example.
- 14.(a) Explain Parallel Database Architectures.  
(b) Explain Network types.
15. Explain how different types of failures are handled by two-phase commit protocol.
16. Explain about Intra operation parallelism.
17. Write short notes on the following:  
(a) ODMG OQL  
(b) Overlapping constraints  
(c) Parallel systems

\*\*\*\*\*

**FACULTY OF ENGINEERING****B.E. 4/4 (CSE) II - Semester (Main & Backlog) Examination, May / June 2019****Subject : Human Computer Interaction (Elective – II)****Time : 3 Hours****Max. Marks: 75****Note: Answer all questions from Part-A & any five questions from Part-B.****PART – A (25 Marks)**

- 1 Define analyzing Interaction paradigms. (2)
- 2 What is internal consistency? (3)
- 3 List out different input devices and explain what is a logical input device. (3)
- 4 What is wizard? (2)
- 5 What are the different Interaction Design models? (3)
- 6 Define different types of boards. (3)
- 7 Classify computer mediated communication. (2)
- 8 Evaluate usability goals. (2)
- 9 Explain about GOMS. (3)
- 10 Define "user- centered approach". (2)

**PART – B (50 Marks)**

- 11 a) Define the business function. What guidelines are used to understand the business function. (5)
- b) Describe the characteristics of Web user interfaces? (5)
- 12 a) What are different Navigation aids in web system? (6)
- b) Explain about Iterative Design. (4)
- 13 Explain briefly about Interface Design Standards. (10)
- 14 Write short notes on
  - a) Writing scripts. (5)
  - b) Running a pilot test. (5)
- 15 Discuss about technical issues Concerning haptics? (10)
- 16 a) Explain about tele operation. (5)
- b) Define "scientific visualization". (5)
- 17 a) Explain what is dual nature of Icons. (4)
- b) Explain about web text and Dynamic text presentation. (6)

\*\*\*\*\*

**FACULTY OF ENGINEERING****B.E. 4/4 (CSE) II-Semester (Main) Examination, April / May 2013****Subject : Multimedia Systems  
(Elective-III)****Time : 3 Hours****Max. Marks: 75****Note: Answer all questions of Part - A and answer any five questions from Part-B.****PART – A (25 Marks)**

1. What are the uses of multimedia? (2)
2. What is Quantization? (3)
3. What is meant by Font? Differentiate between bitmap fonts and vector fonts. (3)
4. Define mesh, curves and patches highlighting their importance in 3D graphics. (3)
5. Differentiate between a tone and a note. (2)
6. What is the difference between (a) Video and motion picture (b) Video and animation (3)
7. What is MMDBs? (2)
8. What is Ripping? (2)
9. What is meant by linear interpolation? (2)
10. What is meant by visual reality? (3)

**PART – B (5x10=50 Marks)**

- 11.(a) Distinguish between multimedia 'production' and 'play back'. How and why would Hardware and software requirements vary in each case? (6)
- (b) Explain Nyquist's sampling theorem and its importance. (4)
- 12.(a) Explain how a time domain signal is converted to frequency domain by forward DCT. (5)
- (b) What is a basis function? Explain the nature of the sine and cosine basis functions in DFT. (5)
- 13.(a) Distinguish between the Huffman coding and LZW coding methods of text compression. (4)
- (b) How does the CMYK color model represent color information? Why is it called a subtractive model. (6)
- 14.(a) Explain how 3D objects can be created from 2D shapes using lathing, extrusion and lofting methods. (5)
- (b) Compare in brief the various video recording systems. (5)
- 15.(a) Explain the steps for creating parameters curve editing and hierarchical animation. (5)
- (b) Give the complete details about MPEG standards. (5)
- 16.(a) Describe the working principle of encoding digital data on a CD surface. (6)
- (b) Differentiate between VCD and super-VCD formats. (4)
17. Write short notes on any **two** of the following: (10)
  - (a) MMDBS
  - (b) SGML and MHEG
  - (c) 3D modeling

**FACULTY OF ENGINEERING****B.E. 4/4 (CSE) II – Semester (Main & Backlog) Examination, May / June 2019****Subject: Simulation & Modeling (Elective – II)****Time: 3 Hours****Max.Marks: 75****Note: Answer all questions from Part A. Answer any five questions from Part B.****PART – A (25 Marks)**

- 1 Define discrete systems.
- 2 Define simulation. Give an example.
- 3 Define queuing systems.
- 4 What is statistical model?
- 5 Define uniform distribution.
- 6 What are pseudo-random numbers?
- 7 Define chi-square test.
- 8 What is KS test?
- 9 What is Output Data Analysis?
- 10 Define stochastic nature of output data.

2  
3  
2  
3  
2  
2  
3  
2  
3  
3

**PART – B (5x10 = 50 Marks)**

- 11 Discuss:
  - a) Advantages and disadvantages of simulation.
  - b) Steps in simulation study.
- 12 Explain:
  - a) GPSS
  - b) SLAM
- 13 Explain:
  - a) Properties of random numbers.
  - b) Poisson distribution.
- 14 Discuss:
  - a) Time series input models.
  - b) Input validation using tuning test.
- 15 Explain:
  - a) Measures of performance and their estimation.
  - b) Output analysis for steady state simulations.
- 16 Discuss:
  - a) Areas of application
  - b) Gamma distribution
- 17 Write short notes on:
  - a) Weibul's distribution
  - b) Verification and validation.

5  
5  
  
5  
5  
  
5  
5  
  
5  
5  
  
5  
5  
  
5  
5

\*\*\*\*

**FACULTY OF ENGINEERING****BE 4/4 (CSE) II Semester (Main & Backlog) Examination, May/June 2019****Subject: Software Quality and Testing (Elective-II)****Time: 3 Hours****Max. Marks: 75****Note: Answer all questions from Part-A & any five questions from Part-B****PART – A (25 Marks)**

- |                                                                 |   |
|-----------------------------------------------------------------|---|
| 1. Define quality and software quality Assurance.               | 2 |
| 2. Differentiate 3 sigma and 6 sigma                            | 3 |
| 3. How to implement software quality metrics?                   | 2 |
| 4. What is function point? How do you compute it?               | 3 |
| 5. Why the defects are hard to find?                            | 2 |
| 6. Distinguish validation and verification                      | 2 |
| 7. Compare glass box testing and black box testing.             | 3 |
| 8. Write short notes on Pareto analysis.                        | 2 |
| 9. Which testing is done on executed code? Give its objectives. | 3 |
| 10. Discuss about work bench concept with an example.           | 3 |

**PART – B (50 Marks)**

- |                                                                                   |    |
|-----------------------------------------------------------------------------------|----|
| 11.a) Define and explain about the steps in software quality Assurance plan.      | 5  |
| b) Discuss about CMM and CMMI.                                                    | 5  |
| 12. What is the need of software quality metrics? Explain In-process metrics.     | 10 |
| 13. How to build a structural approach to software testing?                       | 10 |
| 14. Explain the methodology to evaluate automated testing tools.                  | 10 |
| 15. What is V testing? Explain the work bench used to test client/server systems. | 10 |
| 16.a) Explain eleven steps of testing process.                                    | 5  |
| b) Discuss Load Runner testing tool.                                              | 5  |
| 17. Write short notes on following:                                               |    |
| a. Testing Tactics Checklist                                                      | 4  |
| b. JAD                                                                            | 3  |
| c. Boundary Value Analysis                                                        | 3  |

\*\*\*\*\*